APPENDIX G

ServiceModelEx Catalog

CollectionExtensions

Category
C#

Description
Augments the LINQ collection extensions with additional methods.

See Also
`ArrayExtensions`

ArrayExtensions

Category
C# 3.0

Description
Provides all the LINQ extensions that return a collection except when operating on an array, in which case they return an array. Augments the LINQ collection extensions with additional methods for arrays.

See Also
`CollectionExtensions`

InProcFactory<S,I>

Category
Hosting

Description
Instantiates a .NET class over WCF. No config file or host is required.

See Also
`DuplexClientBase<T,C>`

ServiceThrottleHelper
NetNamedPipeContextBinding
ServiceHost<T>
GenericResolver

WcfWrapper<S,I>

Category
Hosting

Description
Wraps the InProcFactory<S,I> to provide regular .NET syntax for creating and managing the service. The result is indistinguishable from plain .NET programming, yet the object is instantiated and called over WCF.

See Also
InProcFactory<S,I>
NetNamedPipeContextBinding

ServiceHost<T>

Category
Hosting

Description
Adds type safety to ServiceHost, adds support for string-based base addresses, streamlines and automates metadata publishing, automates handling known types, adds any error handler extensions, adds the ErrorHandlerBehavior to the service, including exception details in faults, verifies all queues, purges queues on shutdown in debug mode, enables security audits, and reads throttle values.

See Also
GenericResolver
LogbookManager
LogbookManagerClient
ErrorHandlerHelper
DebugHelper
QueuedServiceHelper

MetadataHelper

Category
Contracts

Description
Performs type-safe contract queries of service metadata.

See Also

`ServiceBusMetadataHelper`

DataContractSerializer<T>

Category

Data Contracts

Description

Generic type-safe wrapper of `DataContractSerializer`.

GenericResolver, GenericResolverBehaviorAttribute, GenericResolverInstaller

Category

Data Contracts

Description

Automates handling known types. Can add specific known types or implicitly all types used in the application. Installed using host or proxy extension method, or declaratively on the service class.

See Also

`ServiceHost<T>`
`InProcFactory<S,I>`

ServiceThrottleHelper

Category

Instance Management

Description

Sets throttle extensions for `ServiceHost`.

IInstanceStore<ID,T>

Category

Durable Services

Description

Defines an interface for any instance store.

See Also

`FileInstanceStore<ID,T>`
`TransactionalMemoryStore<ID,T>`

FileInstanceStore<ID,T>, FilePersistenceProvider, FilePersistenceProviderFactory

Category
Durable Services

Description
File-based store; corresponding provider and provider factory.

See Also
`IInstanceStore<ID,T>`
`TransactionalMemoryProviderFactory`
`TransactionalInstanceProviderFactory`

MemoryProvider, MemoryProviderFactory

Category
Durable Services

Description
Abstract provider and provider factory for in-memory store.

See Also
`TransactionalMemoryProviderFactory`
`TransactionalInstanceProviderFactory`
`TransactionalMemoryProvider`
`TransactionalInstanceProvider`

TransactionalMemoryStore<ID,T>, TransactionalMemoryProvider, TransactionalMemoryProviderFactory

Category
Durable Services

Description
In-memory transactional store shared across clients; corresponding provider and provider factory.

See Also
`IInstanceStore<ID,T>`
`TransactionalInstanceProviderFactory`
`FilePersistenceProviderFactory`

TransactionalInstanceStore<ID,T>, TransactionalInstanceProvider, TransactionalInstanceProviderFactory

Category
Durable Services

Description
Per client session in-memory transactional store; corresponding provider and provider factory.

See Also
`IInstanceStore<ID,T>`
`TransactionalMemoryProviderFactory`
`FilePersistenceProviderFactory`

InstanceContext<T>, DuplexClientBase<T,C>, DuplexChannelFactory<T,C>

Category
Operations

Description
Provide type safety for the client with callback contracts, via either a proxy class or a channel factory.

See Also
`SecureDuplexClientBase<T,C>`

DebugHelper

Category
Faults

Description
Constant to include exception details in faults in debug mode only, extension for `FaultException<ExceptionDetail>` to extract the CLR exception.

ErrorHandlerHelper

Category
Faults

Description
Used for promoting exceptions and logging errors to `LogbookManager`.

See Also

`ErrorHandlerBehaviorAttribute`
`CallbackErrorHandlerBehaviorAttribute`

ErrorHandlerBehaviorAttribute, CallbackErrorHandlerBehaviorAttribute

Category
Faults

Description
Used for declaratively promoting exceptions and logging errors to `LogbookManager`; attributes for service and callback.

See Also
`ServiceHost<T>`
`ErrorHandlerHelper`

ILogbookManager, LogbookManagerClient, LogbookManager

Category
Faults

Description
Service contract for logging errors; proxy for the contract; the logbook service itself.

See Also
`ErrorHandlerHelper`

BindingRequirementAttribute

Category
Transactions, General

Description
Used for declaratively enforcing the Client/Service propagation mode on the service or the callback object, declaratively insisting on an intranet-only binding, and declaratively requiring reliability.

ResourceManager, TransactionalLock

Category
Transactions

Description
Helper classes useful when implementing a transactional resource manager.

See Also

`Transactional<T>`

Transactional<T>

Category

Transactions

Description

A generic volatile resource manager. Performs transactions on any serializable type without compromising on the type programming model.

See Also

`ResourceManager`
`TransactionalLock`

TransactionalCollection<C,T>

Category

Transactions

Description

A generic abstract volatile resource manager collection, useful when implementing a custom transactional collection.

See Also

`Transactional<T>`

TransactionalArray<T>, TransactionalDictionary<K,T>, TransactionalLinkedList<T>, TransactionalList<T>, TransactionalQueue<T>, TransactionalSortedDictionary<K,T>, TransactionalSortedList<K,T>, TransactionalStack<T>

Category

Transactions

Description

The array and all the collections in .NET as transactional volatile resource managers. The collections and the array are polymorphic with the nontransactional built-in collections and are completely interchangeable.

See Also

`TransactionalCollection<C,T>`

TransactionalBehaviorAttribute

Category
Transactions

Description
Enables transparent transactional programming on a per-session service. Negates the need to use a per-call service just for transactions.

See Also
`TransactionalMemoryStore<ID,T>`
`TransactionalMemoryProvider`
`TransactionalMemoryProviderFactory`
`Transactional<T>`
`TransactionalDictionary<K,T>`
`NetNamedPipeContextBinding`

AsyncOneWayClientBase<T>

Category
Concurrency management

Description
A proxy helper class that issues true asynchronous fire-and-forget one-way calls. Requires all operations on contract to be one-way.

FormHost<F>

Category
Concurrency and Windows Forms

Description
Turns a Windows Forms form into a service, without affecting the form's implementation. Service calls simply update the UI.

SafeButton, SafeLabel, SafeListBox, SafeProgressBar, SafeStatusBar, SafeTextBox, SafeTrackBar

Category
Concurrency and Windows Forms

Description
Controls that can be accessed by any service or callback call, on any thread, to update the UI.

See Also
`FormHost<F>`

ThreadPoolSynchronizer, ThreadPoolBehaviorAttribute, CallbackThreadPoolBehaviorAttribute, ThreadPoolHelper

Category
Custom Synchronization Context

Description
Custom thread pool synchronization context; attributes to apply it declaratively on the service or a callback object; helper class to close the threads in the pool.

See Also
`AffinitySynchronizer`
`PrioritySynchronizer`
`ThreadAffinityBehaviorAttribute`
`PriorityCallsBehaviorAttribute`

AffinitySynchronizer, ThreadAffinityBehaviorAttribute, CallbackThreadAffinityBehaviorAttribute, HostThreadAffinity

Category
Custom Synchronization Context

Description
Custom synchronization context providing thread affinity; attributes to apply it on the service and the callback; extension methods to apply at the host level.

See Also
`ThreadPoolSynchronizer`
`ThreadPoolBehaviorAttribute`
`CallbackThreadPoolBehaviorAttribute`

PrioritySynchronizer, PriorityClientBase<T>, PriorityContext, PriorityCallsBehaviorAttribute

Category
Custom Synchronization Context

Description
Custom synchronization context executing calls by priority; proxy class to pass priority in headers; helper class to extract priority from headers; attribute to apply priority processing on service.

See Also
`ThreadPoolSynchronizer`
`ThreadPoolBehaviorAttribute`
`GenericContext<T>`

HeaderClientBase<T,H>

QueuedServiceHelper

Category
Queued Services

Description
Used for verifying all queues on the client, verifying a queue for an endpoint as an extension, and purging a queue.

See Also
ServiceHost<T>

ClientResponseBase<T>, ServiceResponseBase<T>, ResponseContext

Category
Queued Services

Description
Framework for supporting a response service. Proxy class for the client to pass a response address and method IDs; a proxy class for the service to automate calling the response service proxy; response context for direct manipulation.

See Also
GenericContext<T>
HeaderClientBase<T,H>
ClientBufferResponseBase<T>
ServiceBufferResponseBase<T>

CredentialsManager, AspNetSqlProviderService

Category
Security and ASP.NET Providers

Description
Smart client application for managing the ASP.NET credentials store; WCF service wrapping the ASP.NET providers.

See Also
MetadataHelper

SecureClientBase<T>, SecureDuplexClientBase<T,C>, SecurityBehaviorAttribute, SecurityHelper, ServiceSecurity

Category
Declarative Security

Description
Declarative security framework. Declarative security proxy class for regular and duplex calls; attribute for declarative support for the service; declarative security extension methods for regular and duplex type-safe channel factories; declarative security extension methods for the host; extension methods for automatic impersonation; enumeration for declarative security scenarios.

See Also
`DuplexClientBase<T,C>`
`DuplexChannelFactory<T,C>`

IServiceBusProperties

Category
Service Bus

Description
A helper interface for providing the service bus credentials and addresses used. All service bus hosts and proxies in ServiceModelEx support `IServiceBusProperties`.

See Also
`DiscoverableServiceHost`
`ServiceBusHost`
`BufferedServiceBusHost<T>`
`ServiceBusEventsHost`
`ServiceBusClientBase<T>`
`ServiceBusDuplexClientBase<T,C>`
`BufferedServiceBusClient<T>`
`ServiceBusEventsClientBase<T>`

ServiceBusHelper

Category
Service Bus

Description
A static helper class for streamlining common operations on the service bus such as extracting the service namespace, administering buffers, and setting the service bus credentials. Used internally by many of the ServiceModelEx classes but can be used directly as well.

ServiceBusHost, ServiceBusClientBase<T>, ServiceBusDuplexClientBase<T,C>

Category
Service Bus

Description
A dedicated host proxy class for services that use the service bus. Automate configuring Message security, with both anonymous and authenticated client credentials. Offers type-safe proxy for duplex callbacks.

See Also
`DiscoverableServiceHost`
`ServiceBusEventsHost`
`ServiceBusEventsClientBase<T>`

DiscoverableServiceHost, IServiceBusDicovery, IServiceBusDiscoveryCallback, ServiceBusDiscoveryClient, ServiceBusDiscoveryFactory, ServiceBusDiscoveryHelper, IServiceBusAnnouncements, ServiceBusAnnouncementSink<T>

Category
Service Bus

Description
A small framework that adds WCF-like discovery support for the service bus. `DiscoverableServiceHost` publishes the service endpoints to the service bus registry, receives client's discovery requests on `IServiceBusDicovery` and responds to the requests on the client's provided endpoint of `IServiceBusDiscoveryCallback`. The client uses `ServiceBusDiscoveryClient` to discover the service. The client can use the helper classes of `ServiceBusDiscoveryFactory` and `ServiceBusDiscoveryHelper` to streamline the interaction. `DiscoverableServiceHost` can also notify its clients about service state, and the client uses the implementation of `IServiceBusAnnouncements` in `ServiceBusAnnouncementSink<T>` to receive the notifications.

See Also
`ServiceBusHost`
`DiscoveryHelper`
`DiscoveryFactory`
`AnnouncementSink<T>`

OneWayClientBase<T>

Category
Service Bus

Description
A dedicated proxy class for the one-way relay binding. Specializes `ServiceBusClientBase<T>` and streamlines the particular issues of configuring Message security for the one-way relay binding.

See Also
`ServiceBusClientBase`

BufferedServiceBusHost<T>, BufferedServiceBusClient<T>, BufferedServiceBusChannelFactory<T>

Category
Service Bus

Description
A small framework for supporting structured calls to and from service bus buffers, instead of using raw WCF messages. Uses the generic interceptor technique along with message headers for extensibility such as a response service.

See Also
`InterceptorClientBase<T>`
`HeaderClientBase<T,H>`
`ServiceBufferResponseBase<T>`
`ClientBufferResponseBase<T>`

ClientBufferResponseBase<T>, ServiceBufferResponseBase<T>

Category
Service Bus

Description
Dedicated proxies to support a response interaction over the service bus buffers. `ClientBufferResponseBase<T>` is used by the client to call the service, and `ServiceBufferResponseBase<T>` is used by the service to call the response service. Both specialize `BufferedServiceBusClient<T>` and automate passing the information in the message headers. Used just like the response service over MSMQ. Mandate the use of `BufferedServiceBusHost<T>` for the buffered service that responds.

See Also
`BufferedServiceBusClient<T>`
`BufferedServiceBusHost<T>`
`ClientResponseBase<T>`
`ServiceResponseBase<T>`
`ResponseContext`

ServiceBusMetadataHelper

Category
Service Bus

Description
Accepts service bus credentials and performs type-safe contract queries of service metadata.

See Also
`MetadataHelper`

HeaderClientBase<T,H>, HeaderChannelFactory<T,H>, GenericContext<T>

Category
Message Headers

Description
Streamlines passing custom out-of-band parameters to the proxy or channel factory; helper class for extracting and setting the information in the headers.

See Also
`InterceptorClientBase<T>`
`InterceptorChannelFactory<T>`

ContextClientBase<T>, ContextManager

Category
Context Bindings

Description
Streamlines passing custom out-of-band parameters to the proxy over the context bindings; helper class for extracting and setting the information in the context and managing standard IDs.

See Also
`GenericContext<T>`
`HeaderClientBase<T,H>`

NetNamedPipeContextBinding, NetNamedPipeContextBindingElement, NetNamedPipeContextBindingCollectionElement

Category
Context Bindings

Description
Adds the context protocol to the IPC binding; helper classes (not required for direct use) for enabling administrative configuration.

See Also

`ContextClientBase<T>`
`ContextManager`
`InProcFactory<S,I>`

DiscoveryHelper

Category

Discovery

Description

Helper classes for finding available base addresses, enabling discovery on the host, discovering service addresses based on cardinality and scope.

See Also

`DiscoveryFactory`
`ServiceBusDiscoveryFactory`
`ServiceBusDiscoveryHelper`

DiscoveryFactory

Category

Discovery

Description

A discovery helper class. It is both a class factory for creating proxies to discovered services and creating discoverable hosts. Can create proxies based on cardinality and scope, or discover the MEX endpoint for address and binding.

See Also

`DiscoveryFactory`
`DiscoveryHelper`
`ServiceBusDiscoveryHelper`

AddressesContainer<T>

Category

Discovery

Description

Elaborate container for endpoint addresses, offers multiple indexes and enumerators.

See Also

`AnnouncementSink<T>`
`DiscoveredServices<T>`

AnnouncementSink<T>

Category
Discovery

Description
Encapsulates an announcement endpoint yet specific to contract type. Allows accessing announced services addresses list via the base class of `AddressesContainer<T>` as well as subscribing to events notifying of services.

See Also
`AddressesContainer<T>`
`DiscoveredServices<T>`
`DiscoveryPublishService<T>`
`ServiceBusAnnouncementSink<T>`

DiscoveredServices<T>

Category
Discovery

Description
Continuously discovers in the background services using the discovery endpoint. Allows accessing discovered services addresses list via the base class of `AddressesContainer<T>`.

See Also
`AddressesContainer<T>`
`AnnouncementSink<T>`
`DiscoveryPublishService<T>`

PersistentSubscriptionServiceClient, IPersistentSubscriptionService, ISubscriptionService, PersistentSubscription, PublishService<T>, SubscriptionManager<T>, PersistentSubscriptionManager

Category
Publish-Subscribe

Description
Framework for implementing publish-subscribe solution. Proxy class for adding persistent subscribers; service contracts for the publishing and subscription services; base classes for the implementation of the publishing and subscription services; demo application for managing persistent subscriptions.

See Also
`MetadataHelper`
`DuplexClientBase<T,C>`

`DuplexChannelFactory<T,C>`

ServiceBusEventsHost, ServiceBusEventsClientBase<T>

Category
Publish-Subscribe and Service Bus

Description
A host specific to the service bus events relay binding that allows a service to subscribe to events, and a matching proxy for the publishing clients. They work in concert to support discrete events using the service bus as an events hub.

See Also
`ServiceBusHost`
`OneWayClientBase<T>`

DiscoveryPublishService <T>

Category
Publish-Subscribe and Discovery

Description
Base class for a discovery-enabled publish-subscribe service. `DiscoveryPublishService<T>` supports a discover endpoint so that publishers can discover it. `DiscoveryPublishService<T>` provides an announcement endpoint for subscribers, and it continuously discovers subscribers with a discovery endpoint. Also provides for helper methods for hosts and publishers.

See Also
`AnnouncementSink<T>`
`DiscoveredServices<T>`

GenericInvoker, OperationInterceptorBehaviorAttribute, ServiceInterceptorBehaviorAttribute

Category
Generic Interceptor

Description
Framework for easy injection of an interceptor to do pre- or post-call processing; attributes for operation- and service-level declarative use.

See Also
`InterceptorClientBase<T>`
`SecurityCallStackInterceptor`
`OperationSecurityCallStackAttribute`

SecurityCallStackBehaviorAttribute

InterceptorClientBase<T>, InterceptorChannelFactory<T>

Category
Generic Interceptor

Description
Proxy base class and channel factory for easy interception of client calls.

See Also
HeaderClientBase<T,H>
OperationInterceptorBehaviorAttribute
ServiceInterceptorBehaviorAttribute

SecurityCallFrame, SecurityCallStack, SecurityCallStackContext, SecurityCallStackClientBase<T>, SecurityCallStackInterceptor, OperationSecurityCallStackAttribute, SecurityCallStackBehaviorAttribute

Category
Security Identities

Description
Framework for propagating the stack of callers' identities in the message headers; definitions of the call stack and supporting types; proxy class enabling the client to push its identity; interceptor for managing the call stack; attributes to inject the interceptor at the operation or service level.

See Also
GenericInvoker
OperationInterceptorBehaviorAttribute
ServiceInterceptorBehaviorAttribute
GenericContext<T>
InterceptorClientBase<T>